

SMS Sewer Maintenance Structures

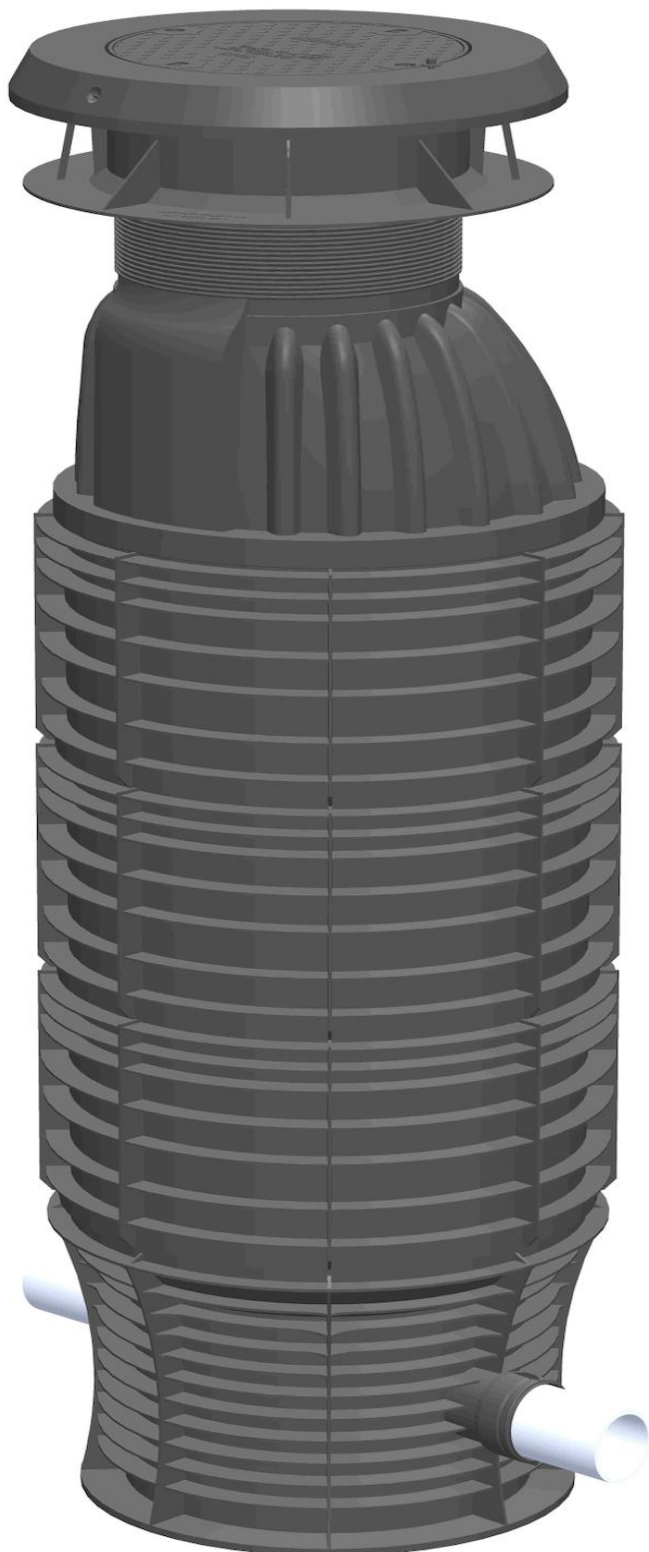


SMS/ROMOLD DN1000 MAINTENANCE HOLES



ROMOLD

SMS/ROMOLD DN1000 MAINTENANCE HOLE



OVERVIEW

The SMS Maintenance Holes are designed for sewer applications with strict quality compliance requirements, for example in terms of density and durability. Furthermore, these systems are designed to allow safe and time-efficient installation.

- We offer a range of maintenance Hole systems for the optimal collection of sewer/wastewater.
- Ease of access for maintenance equipment for pipeline systems (camera's, mirror & cleaning tools) which are permitted underground for inspection and maintenance.
- SMS - Maintenance Hole provide alternate solution to traditional inspection Holes.
- SMS - Maintenance Hole can be installed below roadways and footpaths to an installation depth of 6m (max).
- Easily integrated to sewer networks configuration with diameter of DN150, DN225, DN300, DN375.
- Designed, Engineered and manufactured in Germany;
- Follow all installation guides including minimum gap allowances to ensure any loads are not directly transferred to the cone.

BENEFITS

- No Contamination of Soil and Groundwater
- Easy installation as each element is no more than 15-75kg.
- Reduction of mechanical equipment with quicker installation over traditional Holes.
- Flat bottom of the SMS Maintenance Holes provides a controlled stable platform to work with.
- Coupling sleeves allow minimum +/- 3.5 degrees for pipeline adjustments.
- Chemical-resistant chambers, suitable for industrial wastewater & sewer.
- SMS Maintenance Holes are Resistance to Traffic Loads with specially developed top hats/access cover arrangements to facilitate distribution of the load requirements.





MAINTENANCE HOLE BASE CONFIGURATIONS

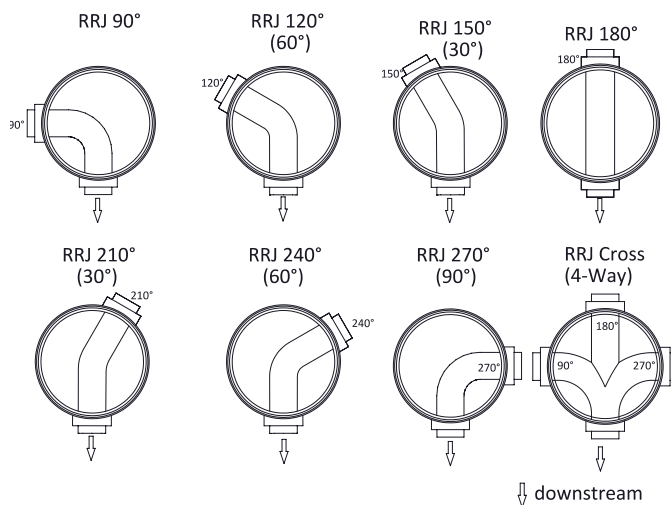


Table 2 - SMS DN1000 Maintenance Hole Configurations

Product Code	Configuration	Inlet/Outlet Size
MHB1000150090	RRJ 90°	DN150
MHB1000150120	RRJ 120°	DN150
MHB1000150150	RRJ 150°	DN150
MHB1000150180S	RRJ 180° Straight	DN150
MHB1000150210	RRJ 210°	DN150
MHB1000150240	RRJ 240°	DN150
MHB1000150270	RRJ 270°	DN150
MHB10001504WC	RRJ Cross (4-Way)	DN150
MHB1000225090	RRJ 90°	DN225
MHB1000225120	RRJ 120°	DN225
MHB1000225150	RRJ 150°	DN225
MHB1000225180	RRJ 180° Straight	DN225
MHB1000225210	RRJ 210°	DN225
MHB1000225240	RRJ 240°	DN225
MHB1000225270	RRJ 270°	DN225
MHB10002254WC	RRJ Cross (4-Way)	DN225
MHB1000300090	RRJ 90°	DN300
MHB1000300120	RRJ 120°	DN300
MHB1000300150	RRJ 150°	DN300
MHB1000300180S	RRJ 180° Straight	DN300
MHB1000300210	RRJ 210°	DN300
MHB1000300240	RRJ 240°	DN300
MHB1000300270	RRJ 270°	DN300
MHB10003004WC	RRJ Cross (4-Way)	DN300
MHB1000375090	RRJ 90°	DN375
MHB1000375120	RRJ 120°	DN375
MHB1000375150	RRJ 150°	DN375
MHB1000375180S	RRJ 180° Straight	DN375
MHB1000375210	RRJ 210°	DN375
MHB1000375240	RRJ 240°	DN375
MHB1000375270	RRJ 270°	DN375
MHB10003754WC	RRJ Cross (4-Way)	DN375

SPECIFICATIONS & REQUIREMENTS

Table 1 - SMS DN1000 Maintenance Hole (MH)

Riser Nominal Dia (DN)	DN1000
100% Recyclable	Yes
Person Entry	Yes
Chemical Resistance	pH 1-13
Base Material	Polypropylene (PP)
Reducing Cone Material	Polypropylene (PP)
Riser Element Material	Polypropylene (PP)
Gasket Material	(Ethylene propylene Diene Monomer EPDM
Max Installation Depth	6
Max Ground Water Height	5m above inflow pipe invert level
Joint Grade & Deflection Angle	Minimum (Vert & Hor) $\pm 3.75^\circ$
Max Traffic Load	Class D to AS3966 "Access Covers and Grates"
Pipe Inlet & Outlet Material	Drainage Waster & Vent (DWV) to AS/NZ1260
Pipe Inlet & Outlet Diameter	DN100, DN150, DN225, DN300 & DN375
High Level Inlet Connection	DN 150 Cut-in Connection
Product Certification	Globalmark - WSA137
WSAA Product Appraisal	PA 1317 – Issue 4
WSAA Product Specification	PS-340 Maintenance Holes (MH) (PP) for Non-Pressure Applications - Gravity Sewerage
Relevant Standards	WSA137, EN13598-2, AS3996, AS1646
Service Life	100 Years

The products shown form part of SMS's continuous improvement program and as such the product designs, specifications and materials may change without notice. All warranties relating to accuracy, completeness, or suitability for any particular purpose and all liability for any loss, damage or costs incurred relating to the use of this information are excluded.

ROMOLD

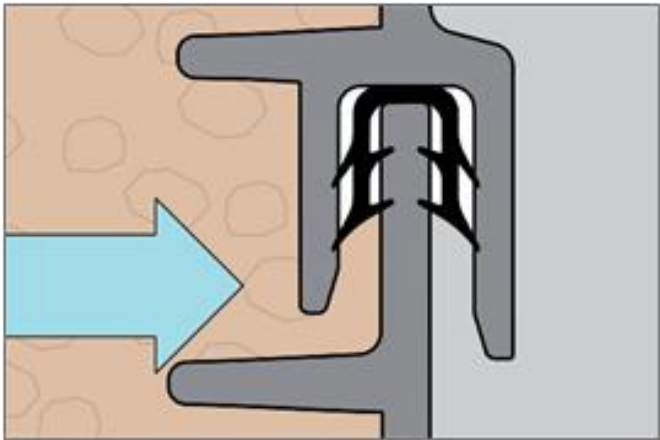
SMS/ROMOLD DN1000 MAINTENANCE HOLE

ADDITIONAL FEATURES - SMS/ROMOLD DN1000 MAINTENANCE HOLE SYSTEM

- Space saving, ideal for inner city areas;
- Optimised stability;
- Height adjustment to the nearest centimeter;
- Corrosion free and anti-slip steps;
- Improved rib spacing;
- Unique and well-proven triple-safety-seal;
- Light grey anti-slip benching;
- Flexible sockets on inlet and outlet;
- Hydraulically optimised channel;
- Over 2 million chamber elements sold;
- More than 30 years of experience in developing;
- Plastic chambers;
- Floatation protection;
- Seamless bends ensure hydraulically optimized channel, no segmentation.



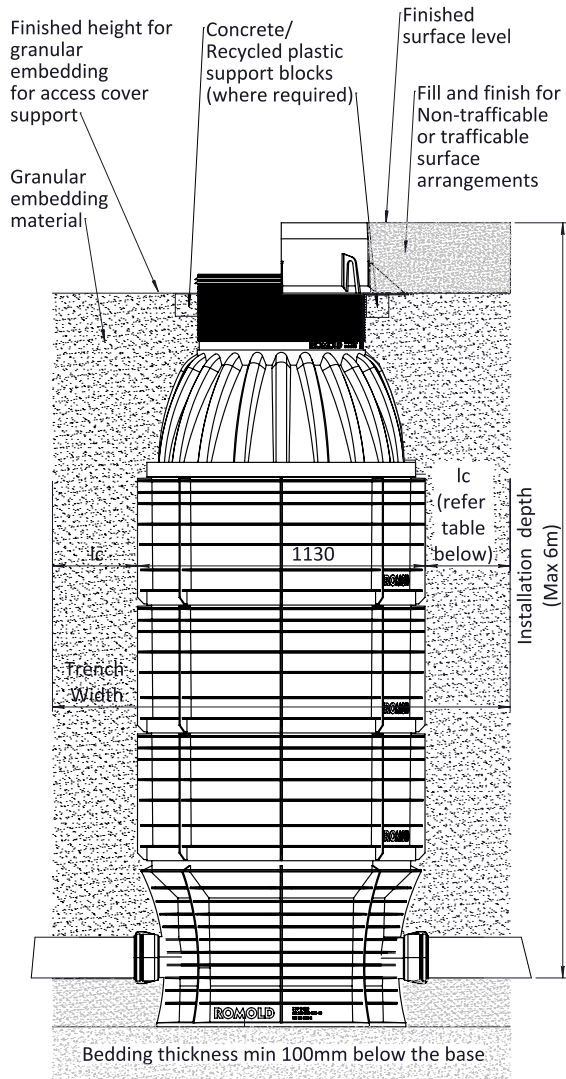
ELEMENT SEAL SYSTEM



3-sided element seal (Triple-Safety-Seal) for connecting the Maintenance Hole elements.

INSTALLATION OVERVIEW

General overview of DN1000 Maintenance Hole utilizing DN1000 Top hat. For installation of different installation options, contact SMS for detailed installation guides.



- Installation notes:**
- Refer to local Water Agency specifications for other specific installation guidelines.
 - For installation in reactive soils, refer to local water agency installation guidelines.
 - Concrete used for installation or concrete components to comply to local water agency specifications.
 - Test installed maintenance hole to local water agency requirements.
 - Refer to AS/NZS 2566 for Buried Flexible Pipelines and local authority requirements for any specific installation requirements.

Bedding, Backfilling/Embedment material:

- Non-cohesive material rounded gravel up to 32mm broken & gravel up to 16mm width.
- Compaction: DPR ≥ 97%.

Trench widths	lc (Min.)	Trench Width (Min.)
Standard Installation	400	1930
Hi Water Table/ Groundwater conditions	500	2130

- Installation depth notes:**
- Standard Installation max 6m depth, in hi water table/ groundwater conditions max 5m depth of installation.

Figure 1 – Typical Installation using Top Hat

MAINTENANCE HOLE CONFIGURATION OPTIONS

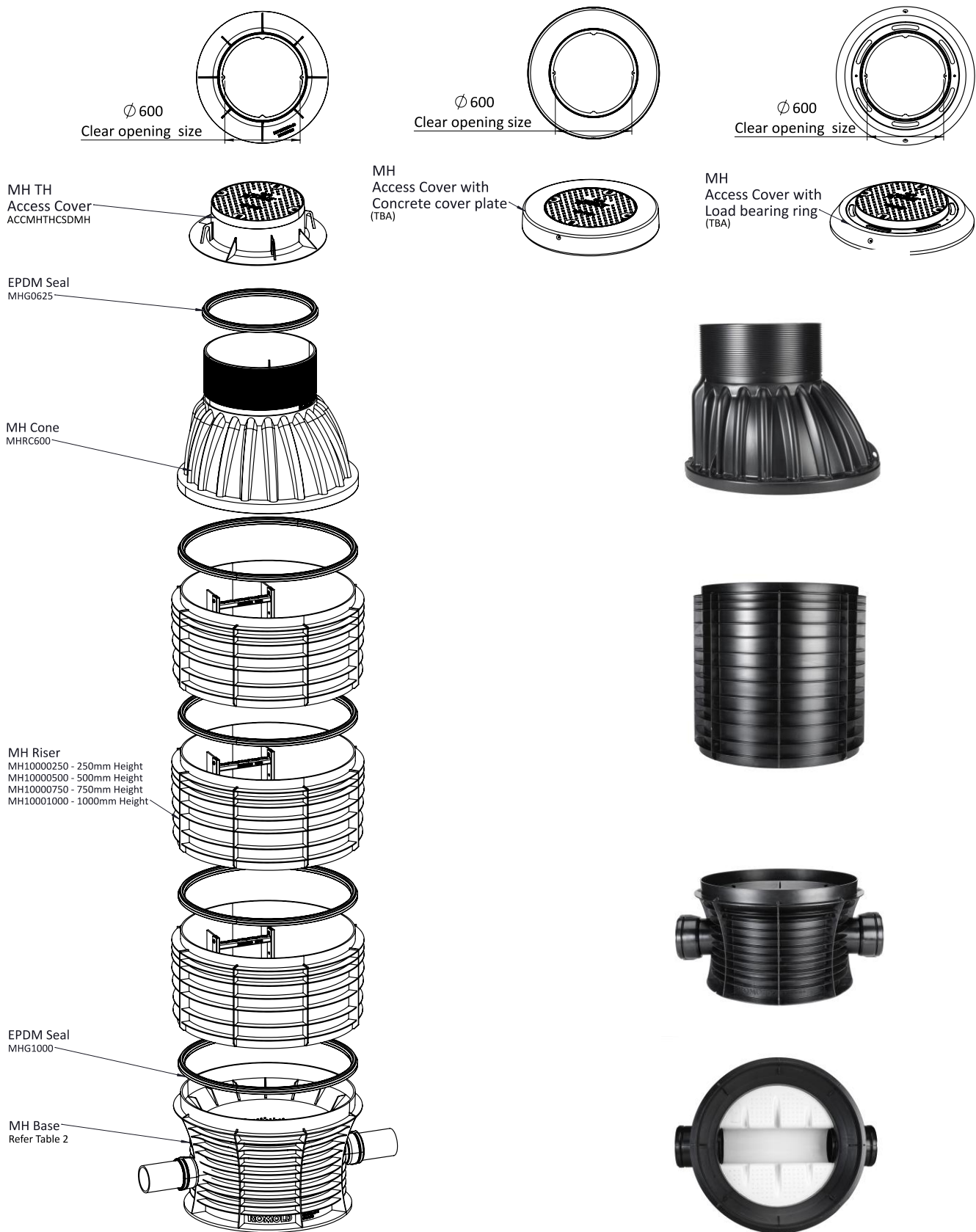


Figure 2 – Typical Installation options

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SYSTEM COMPONENTS

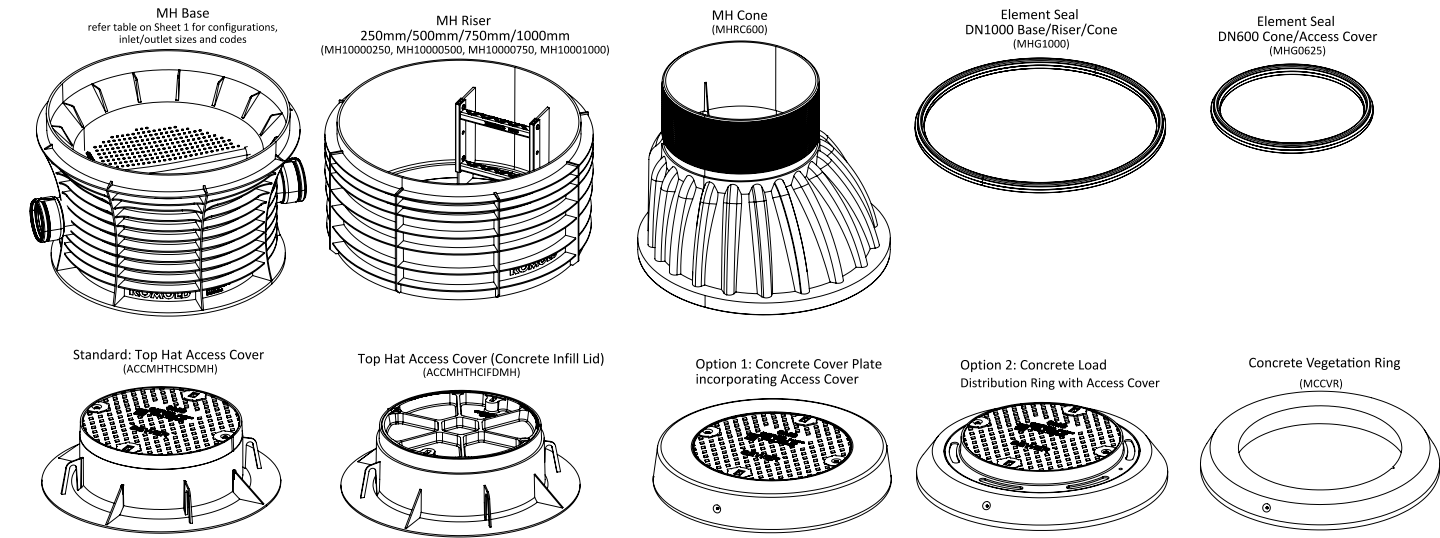


Figure 3 – Typical System Components



Contact SMS for additional configuration detail to meet project specifications.

INSTALLATION HEIGHT CALCULATION & COMPONENT TABLE

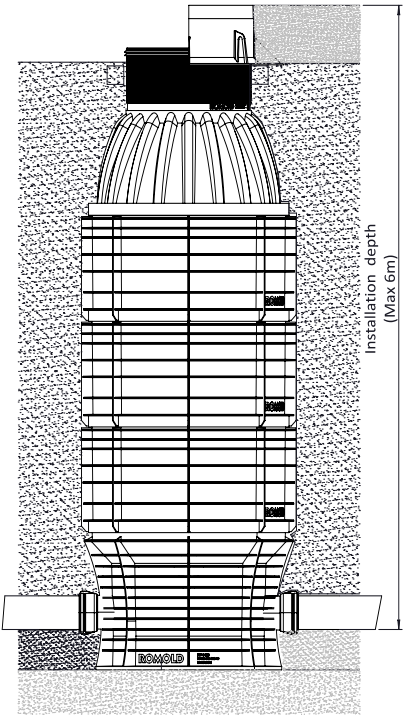


Table 3: Access Cover Top hat Installation Depths

Installation depth		1500	1750	2000	2250	2500	2750	3000	3250	3500	3750	4000	4250	4500	4750	5000	5250	5500	5750
range ->		1750	2000	2250	2500	2750	3000	3250	3500	3750	4000	4250	4500	4750	5000	5250	5500	5750	6000
Component	Product Code	Qty																	
Base	Refer table 2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Riser (250)	MH10000250	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0
Riser (500)	MH10000500	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1
Riser (750)	MH10000750	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0
Riser (1000)	MH100001000	0	0	0	1	1	1	1	2	2	2	2	3	3	3	3	4	4	4
Cone	MHRC600	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
DN600 MH Top hat	ACCMHTHCSD	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Access Cover	MH	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Table 4: Access Cover Option 1 & 2 Installation Depths

Installation depth		1420	1670	1920	2170	2420	2670	2920	3170	3420	3670	3920	4170	4420	4670	4920	5170	5420	5670	5920
range ->		1670	1920	2170	2420	2670	2920	3170	3420	3670	3920	4170	4420	4670	4920	5170	5420	5670	5920	6170
Component	Product Code	Qty																		
Base	Refer table 2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Riser (250)	MH10000250	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0
Riser (500)	MH10000500	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0
Riser (750)	MH10000750	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1
Riser (1000)	MH100001000	0	0	0	1	1	1	1	2	2	2	2	3	3	3	3	4	4	4	4
Cone	MHRC600	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
DN600 MH Access	TBA	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Cover option 1 or 2																				

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